ECON-4210
Economics of Climate Change*

Prof. Casey Wichman

Georgia Institute of Technology

Fall 2026

Course website: `https://www.caseyjwichman.com/econ4210/`
Course meeting time: Mondays and Wednesdays, 2:00–3:15 PM
Classroom: Van Leer E283
Office hours: TBA
Teaching Assistant (TA): TBA
TA Office Hours: TBA

Course description: This course will explore the economic causes and consequences of climate change—the "mother of all externalities"—and potential solutions. Misguided market forces, via unpriced and unregulated greenhouse gas (GHG) emissions, contribute to global warming. Because GHG emissions stem from economic decisions we make every day—from driving to work to powering our computers to eating a hamburger—the economic impact of policies aimed at reducing GHG emissions will be pervasive. In this course, we will use the tools of economics to understand the origins of climate change; to explore the effects of climate on different facets of the economy now and in the future; and to evaluate the effectiveness, costs, and benefits of policies designed to reduce climate change. Particular focus will be given to:

1. Understanding the methods used to measure and value damages from climate change
2. Analyzing the pros and cons of market-based policies designed to reduce GHG emissions
3. Investigating the role of adaptation and innovation as potential climate solutions
4. Evaluating the equity implications of climate impacts and climate policy

Students will acquire a broad understanding of the costs of unchecked climate change and an ability to evaluate current policy debates through an economic lens.

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1 Course Overview

1.1 Course Format

This course will be in-person and, naturally, synchronous. Throughout the semester, we may meet via Zoom to, e.g., host a guest lecturer virtually. These virtual meetings will be the rare exception rather than the rule. Any virtual sessions will be clearly communicated.

1.2 Learning Goals

The following is a list of competencies that I expect students to walk away from this course with:

1. Understand the (basic) science behind human-induced climate change along with the uncertainties involved.
2. Demonstrate the ability to interpret the measurement and scope of climate change damages.
3. Understand the rationale for and analyze the implications of government intervention as a potential solution to market failures and global externalities.
4. Be comfortable applying social cost of carbon to estimate the impacts of a policy change, especially for the estimation of benefits of reducing climate pollution.
5. Understand the basics of risk and uncertainty, including tipping points, fat tails, and catastrophic risk, and be able to communicate the meaning of those terms.
6. Analyze policy recommendations using knowledge about instrument choice and recognition of the distributional impacts. Make recommendations where necessary to account for environmental justice concerns and for political feasibility.

1.3 Prerequisites

Formally, Principles of Macroeconomics (ECON 2105) AND Principles of Microeconomics (ECON 2106) OR Economics and Policy (ECON 2100) are required. Informally, knowledge of basic statistics, calculus, and econometrics will be valuable for understanding readings and concepts at a deeper level, but not expected.

2 Course Expectations

2.1 Attendance

Attendance is expected and will be necessary to succeed in the course. Participation in class discussions will be strongly encouraged. Ask questions. If you do not understand something in class, you are almost certainly not the only one. If you never volunteer an incorrect answer, you aren't taking enough risks.

2.2 Technology policy

You are welcome to use a laptop or tablet to take notes during class. However, I expect you to use your device for class purposes only. Surfing the internet or checking social media during class is distracting to you and to those around you (including the instructor). If I notice that you are not paying attention, I will ask you to put your device away. Wearing headphones or earbuds is not permitted during class.

There is evidence that students learn better when they take notes by hand rather than on a laptop (see, e.g., `https://www.nytimes.com/2017/11/22/business/laptops-not-during-lecture-or-meeting.html`)

2.3 Course website and Canvas

The course schedule, slides, and journal-article readings live on the course website: `https://www.caseyjwichman.com/econ4210/`. The website is the current version of the schedule and will be updated as the semester goes on.

Canvas is used for assignments, grades, book-chapter readings, and course announcements. Beyond course materials, I often link to podcasts, news articles, and other forms of media related to topics covered in the course (or on climate change more generally).

2.4 Email Policy

The safest bet is to send me a message through Canvas. That way, my course inbox stays separate from my occasionally messy email inbox. I am typically quite slow to respond to messages on weekends.

2.5 Grading

There will be no special extra credit or extra work of any kind for the purpose of raising a grade during or after the course. This is to ensure that everybody has equal opportunities to earn their grade and that grades are based on work during the course.

Neither exam grades nor course grades will be curved.

The final grade is calculated as follows (assignments are explained in more detail below):

Attendance & Participation	20 pts
Problem sets ($n = 4$, 5 pts each)	20 pts
In-class exams ($n = 4$, 20 pts each, lowest dropped)	60 pts
Total	100 pts

Final grades are determined using the standard scale:

Course grade:	A	90–100%
	B	80–89.99%
	C	70–79.99%
	D	60–69.99%
	F	0–59.99%

Final grades are not rounded up. Hence, 89.99 is a B not an A. If you are taking this class pass or fail, a grade of C or higher is a passing grade. If you earn a D or an F, you will be given a failing grade for the course.

2.6 Academic integrity

Cheating is unacceptable. You are hereby reminded that you have pledged to uphold the honor code as follows:

> *Having read the Georgia Institute of Technology Academic Honor code, I understand and accept my responsibility as a member of the Georgia Tech community to uphold the Honor Code at all times. In addition, I understand my options for reporting honor violations as detailed in the code.*

Should you be caught cheating in this class you will be prosecuted according to the honor code and policies and procedures established by the Honor Advisory Council. Should you have any questions about this do not hesitate to contact me.

2.7 Course guidelines on using AI tools

The use of AI tools (e.g., ChatGPT) is permitted in this course. However, I *strongly* encourage you to use AI tools as a complement to, rather than substitute for, learning. Writing something out on your own forces you to think through the concepts and ideas in a way that AI tools cannot. If you outsource writing to AI, you outsource thinking to AI as well. Submitting AI-generated output as your own work is in violation of Georgia Tech's Acadmic Honor code, and if I suspect that you have done so, I will reserve right to grade assignments accordingly.

2.8 Office Hours

I will hold "communal" office hours during a set time each week. These office hours are time I set aside each week to make myself available for students. You can visit my office hours with questions about course material, to discuss economic policy issues in the news, future course selection at Tech, fun hikes you've done around Atlanta, travel plans, and anything else related or unrelated to the course. To discuss something privately, you can request a short meeting with me via Canvas.

2.9 Student-Faculty expectations agreement

At Georgia Tech, it is important to strive for an atmosphere of mutual respect, acknowledgment, and responsibility between faculty members and the student body. See `http://www.catalog.gatech.edu/rules/22/` for an articulation of some basic expectation that you can have of me and that I have of you. In the end, simple respect for knowledge, hard work, and cordial interactions will help build the environment we seek. Therefore, I encourage you to remain committed to the ideals of Georgia Tech while in this class.

2.10 Accommodations for students with disabilities

If you are a student with learning needs that require special accommodation, contact the Office of Disability Services at (404) 894-2563 or `http://disabilityservices.gatech.edu/`, as soon as possible, to make an appointment to discuss your special needs and to obtain an accommodations letter. Please also e-mail me as soon as possible to set up a time to discuss your learning needs.

2.11 Why is this syllabus so long?

This syllabus is intended to be a contract in addition to serving as a guide to the course. I promise to follow-through and implement the course as described, so as to set clear expectations for instruction. There should be no surprises. That said, throughout the course, circumstances may change that will require changes to the syllabus. These will be communicated clearly and I will make edits to the syllabus, if necessary, as the semester goes on.

3 Details on course assignments

- Attendance & Participation (20 pts)
  - Attendance and participation is expected.
  - There are 24 regular course meetings, excluding the four in-class exams and GT holidays. Everyone gets 4 "freebies" (no questions asked) – so, attending and participating in 20 lectures gets you the full 20 pts. That translates to 1 pt per class with a cap at 20 points. Attending 16 classes gets you 16 points.
- Problem sets ($n = 4$, 5 pts each, 20 pts total)
  - Problem sets will force you to apply some of the concepts we've talked about in the course. They will vary in format and might involve demostrating how standard economic tools can solve externality problems or they might involve providing an economic argument for/against a discussion about climate change in the news.
  - You may work on problem sets with others in the course, although you will be required to submit your own assignment.
  - You will be given $\sim$1 week to complete each problem set. Details and due dates will be forthcoming during the semester.
- In-class exams ($n = 4$, 20 pts each, lowest score dropped, 60 pts total)
  - Exams are taken in class, on paper. Each exam covers one unit of the course and is not cumulative.
  - Exam dates: Wednesday, Sep. 23 (foundations: externalities, benefit-cost analysis, and discounting); Wednesday, Oct. 14 (measuring and valuing climate impacts); Wednesday, Nov. 11 (climate policy); and Monday, Dec. 7 (uncertainty, geoengineering, international cooperation, and behavior). The exact classes covered by each exam are listed on the course website. There is no exam during finals week.

  - Your lowest exam score is dropped; your three highest scores count 20 points each. Because the lowest score is dropped, there are no makeup exams – a missed exam is the dropped exam. If you have a valid, documented excuse for missing a second exam, contact me before the exam date.

- ⚠ Late assignments will receive a zero without a valid excuse. ⚠

- ⚠ For reasons of fairness, there will be no extra credit. ⚠

4 Overview of course topics

The course will be split into two major sections. Part I will focus on the economic implications of climate change, how economist measure and value climate change impacts, and how those climate "damages" are accounted for in a benefit-cost analysis framework. Part II will focus on approaches to solving climate change from an economic perspective. We will discuss optimal climate policy—including its advantages, drawbacks, and distributional consequences—as well as adaptation strategies, unintended consequences, uncertainty, and the role of international cooperation.

Topics covered in this course are listed below alongside the bigger picture questions each topic addresses.

Bigger Questions	**Detailed Topic**
Part I: Causes and Consequences of Climate Change	
What is climate change?	Climate science 101
How are the climate and the economy linked?	The economy and the climate
Why is climate change an economic problem?	Externalities and public goods
What are the costs and benefits of climate policy?	Benefit-cost analysis
Should I care about my grandchildren's future?	Discounting benefits and costs
Can we put a price on climate impacts?	Measuring/valuing climate impacts and the social cost of carbon
Who bears the burden of climate impacts?	Equity & climate impacts
Part II: Economic Approaches to Solving Climate Change	
How would an economist solve climate change?	Optimal climate policy
Who pays for a carbon tax?	Equity & climate policy
Does behavior undermine climate policy?	Unintended consequences
Can't we just live with climate change?	Adapting to climate change
What don't we know about climate change?	Uncertainty, fat tails, & tipping points
Can we fix climate change with this weird trick?	Geoengineering & offsets
Can we address climate change on our own?	International cooperation
Is changing our own behavior the solution?	Behavioral change

Readings

This course will be based heavily on readings from books and academic journal articles. Readings for each course meeting are listed on the course website (`https://www.caseyjwichman.com/econ4210/`). You will be expected to perform these readings before the listed class date. These readings may be updated over the course of the semester. Journal articles are posted on the course website; book chapters are posted on Canvas. Some articles will be more technical than others. Don't worry if you don't understand every detail. This is normal.

We will rely heavily on readings from the first book listed below. You can access .pdf copies from the Georgia Tech Library using the embedded links below. You are more than welcome to purchase these books if you would prefer a hard copy. Each of them can be found online for reasonable prices. That said, all readings will be provided electronically—-there is no need to purchase anything for this course.

- **[Keohane & Olmstead:]** Keohane, Nathaniel O. and Sheila M. Olmstead, *Markets and the Enviroment*, Island Press, Second Edition, 2016. (Access provided via Georgia Tech Library)

A few more useful and interesting books that we will have readings from (I will provide these readings in PDF form; they are available via the Georgia Tech Library):

- **[Pindyck:]** Pindyck, Robert S., *Climate Future: Averting and Adapting to Climate Change*, Oxford University Press, 2022.
- **[Nordhaus:]** Nordhaus, Willam, *The Climate Casino: Risk, Uncertainty, and Economics for a Warming World.* Yale University Press, 2013. [Link (access provided via Georgia Tech Library)]
- **[Wagner & Weitzman:]** Wagner, Gernot and Martin L. Weitzman, *Climate Shock: The Economic Consequences of a Hotter Planet,* Princeton University Press.
- **[Kahn:]** Kahn, Matthew, E. *Adapting to Climate Change: Markets and the Management of an Uncertain Future*, Yale University Press, 2021.
- **[American Climate Prospectus:]** *American Climate Prospectus: Economic Risks in the United States*, 2014. [link]. (Think of this as a "coffee table book" highlighting and visualizing research on climate impacts.)

Course Schedule

The class-by-class schedule—dates, topics, exams, slides, and the reading list—lives on the course website: `https://www.caseyjwichman.com/econ4210/`. The website is the single, current version of the schedule and is updated as the semester goes on. This document does not repeat the schedule, so the two cannot drift apart.